Option D – Energy and the Environment

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)	Weak force interaction	1			1		
		(ii)	LHS: 2.01456 u and RHS: 2.01410 u **or** $\Delta m = 4.6 \times 10^{-4}$ u (1) E = 0.428 M[eV] **or** 6.85×10^{-14} [J] (1)		2		2	2	
	(b)	(i)	Solar power received per unit area [crossing a plane] accept intensity (1) At right angles to the line joining the earth to the sun, just outside the earth's atmosphere (1)	2			2		
		(ii)	Not 'constant' due to varying solar power output/Earth's elliptical orbit			1	1		
		(iii)	Stefan-Boltzmann **OR** Stefan **OR** Inverse square law (1) $P = \sigma A T^4 = 5.67 \times 10^{-8} \times 6.1 \times 10^{18} \times 5\,780^4 = 3.9 \times 10^{26}$ (1) $I = \frac{P}{A} = \frac{P \textbf{ ecf}}{4\pi \times (1.5 \times 10^{11})^2}$ (1) Answer = 1 379 or 1 365 [W m^{-2}] (1)	1	1 1 1		4	3	
	(c)		Use of Wien's law to show surface emits λ_{peak} of 10 μm (accept 5 - 20 μm range) (1) Partially absorbed by water with reference to values from graph (1) Also absorbed by CO_2 with reference to values from graph (1) Re-emitted in all directions / towards surface [warming Earth] (1)			4	4	1	

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				AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)	**Any 2 × (1) from:** - U235 fissile whereas U238 is not (1) - Only small amount (0.7 %) of U235 in natural uranium (1) - U238 absorbs neutrons	2			2		
		(ii)	Use of $v = r\omega$ to obtain $\omega = 2\,333$ [rad s^{-1}] (1) 371 [rev s^{-1}] (1) Award 1 mark for answer of 186		2		2	2	
		(iii)	$0.7 \times 1.15^n = 5.0$ (1) n = 14.1 resulting in <u>15</u> stages required (1)		2		2	2	
			Question 10 total	**6**	**9**	**5**	**20**	**10**	**0**